Jonah Cha

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Education

University of California, Berkeley · B.S. Electrical Engineering & Computer Science · GPA 4.00 — Expected May 2028

Coursework: Machine Learning (A+), Data Structures (A+), Structure & Interpretation of Computer Programs (A), Discrete Math & Probability (A); Deep Learning and Reinforcement Learning (audited); Computer Vision, Computer Architecture (in progress)

Experience

UCSF · Researcher, two projects — Jan 2026 – present

- Brain decoding: reconstructing images and video from fMRI with a small fMRI→CLIP map driving frozen diffusion generators, across four public datasets.
- Held-out 2-way identification 0.955 across all 8 NSD subjects; top-1 retrieval 155× chance among 897 unseen images; found and fixed a preprocessing bug that had scrambled image–scan pairs.
- Clinical radiology: deep learning and quantitative imaging for early detection and risk stratification.

UC Berkeley · Researcher — Aug 2026 – present

- Foundation models for RNA: transformer architectures and representation-learning objectives for sequence-to-structure prediction, with compute optimisation on AWS Trainium.

Mercor · Machine Learning Engineer — Sep 2025 – Aug 2026

- Post-training for AI agents on stealth projects: reinforcement learning for agents, training data, and evaluations of agent behaviour.

Berkeley AI Research · Researcher, Dawn Song's group — Sep – Dec 2025

- Co-author (10 of 16) of **CyberGym-E2E**, ICML 2026: an end-to-end benchmark of AI agents finding, proving and patching 920 real-world vulnerabilities across 139 open-source C/C++ projects. arxiv.org/abs/2606.04460

Stoa USA · Software Engineering Intern — Jun 2024 – Aug 2025

- Built StoaHub, the national web platform of the Stoa speech and debate league (100,000+ user entries, 50+ tournaments a year), including its cross-tournament rankings. stoahub.org

Uplevel Ops · AI Consultant — May 2024 – Aug 2025

- Cloud AI agents for a legal consultancy's clients: NDA drafting, contract redlining, and invoice review against billing guidelines; Uplevel estimated over $2 million in client value.
- Invoice review on AWS (API Gateway, Lambda, events, S3) with Pinecone retrieval and a strict review prompt: a 100-page invoice reviewed in under a minute.

Projects

Failure detection with a world model · negative result — 2026

- A probe on frozen encoder features localises robot-policy failures (AUROC 0.88–0.92), but no deployable search repaired them better than a random edit of the same size. github.com/MERC13/failure-detection-WM

Steering a frozen π0.5 by its instructions · audit — 2026

- 644 simulated LIBERO episodes over 74 prompts: rephrasing barely moves success and only 2 of 8 tasks show headroom over the original instruction. github.com/MERC13/openpi

AttentionRetention · with Ben Hung and Lucas Zheng — Jun 2026

- Live-lecture tutor: Deepgram transcription, Redis memory and three Claude agents that detect topic boundaries and quiz the student, traced in Arize. github.com/20ben/attentionretention

Deep-learning and scikit-learn builds — 2025

- nanoGPT-style GPT-2 with DDP and a HellaSwag eval, a hand-written DQN, LoRA fine-tuning, a DCGAN, transformer fantasy-football forecasts, and classical ML models. github.com/MERC13/Deep-learning

Skills & awards

Languages	Python, C++, Java, JavaScript, PHP, SQL, HTML/CSS
ML & infra	PyTorch, JAX, HuggingFace, AWS (Lambda, S3, Trainium), Docker, Linux, Git, Redis, Pinecone
Agents	LangChain, LangGraph, LangSmith, AutoGen, MCP, Claude
Awards	SAT 1600/1600, PSAT 1520/1520, AP Scholar with Distinction, 2× National Gold Medal Debater, Presidential Scholar Candidate